

Program Outcomes

Data Analyst with Generative AI Skills

Build AI-Powered Data Solutions & Intelligent Dashboards

Develop Predictive & Machine Learning Models

Automate Insights using Generative AI

Launch Data-Driven AI Products & Startups

Entrepreneurial & Analytical Mindset

About TECHU

TECHU is a future-focused technology skilling platform dedicated to building industry-ready professionals and tech entrepreneurs.

We specialize in hands-on training across Data Analytics, Machine Learning, and Generative AI, with strong emphasis on real-world datasets, business problem solving, and AI-powered decision systems.

Our programs are designed to bridge the gap between traditional data skills and next-generation AI innovation.

Our Faculty

TECHU's faculty comprises experienced data analyst, AI practitioners, analytics consultants, and startup mentors.

They bring real-time industry exposure, practical insights, and project-driven learning to help learners build job-ready, research-ready, and startup-ready data skills.

Frequently Asked Questions (FAQ)

1. What are the prerequisites for this program?

Basic computer knowledge is sufficient. No prior coding experience is mandatory. The program starts from fundamentals of data analysis and gradually advances to Machine Learning, and Generative AI applications.

2. Do you provide placement assistance?

Yes. We provide placement assistance through resume preparation, portfolio building, mock interviews, real-world project exposure, and industry referrals. Learners are also guided toward internships and analytics roles.

3. Is this program suitable for entrepreneurs and startups?

Absolutely. This program is ideal for entrepreneurs who want to build data-driven products, AI-powered dashboards, analytics tools, or SaaS platforms using real-time insights and automation.

WHY TECHU ?

1. Industry-Aligned Curriculum

Designed to meet current industry requirements in Data Analytics, Data Science, Machine Learning, and Generative AI.

2. Hands-On Learning Approach

Learners work on real-world design, marketing, and AI automation projects rather than just theory.

3. Data + AI Integration

Learn how to integrate analytics, ML models, and Generative AI into real business workflows.

4. Generative AI for Data Professionals

Automate reports, build AI-driven insights, create data storytelling tools, and deploy intelligent assistants.

5. End-to-End Data Lifecycle Coverage

From data collection and cleaning to modeling, visualization, and AI deployment.

6. Entrepreneur-Focused Training

Encourages product thinking, analytics consulting skills, and AI startup development.

7. Live Projects & Capstone

Each learner builds portfolio-ready dashboards, ML models, and a final AI-powered data product.

8. Experienced Industry Faculty

Led by professionals with real-time analytics, AI, and consulting experience.

9. Placement Assistance

Support includes resume building, GitHub portfolio preparation, interview training, and job guidance.

10. Flexible Learning Modes

Available in online, offline, and hybrid formats.

11. Post-Training Mentorship

Continued guidance for analytics career growth, certifications, freelance consulting, and startup support.

Our Address

101, Jain Sadguru Images Capital Park, Beside Rameshwaram Cafe, Madhapur, Hyderabad, Telangana 500081

Phone

+91 9000144281

Email

info@techu.in

JGI JAIN CENTER FOR SKILLS

TU TechU INNOVATION LABS PVT LTD

DATA ANALYTICS AI/ML+GEN AI

Duration:
5 Months

Mode:
Online / Hybrid / Offline

10,000+ Professionals Trained Globally

DATA ANALYTICS + AI/ML – COURSE CURRICULUM

Module 1:

Advance Excel

- Functions
- Conditional Statements
- Data Cleaning (Excel)
- Dashboard Building

- Lookups and Pivot Tables
- Macros and Goal Seek analysis

Module 2:

Python Programming Fundamentals

- Introduction to programming
- Python syntax and structure
- Variables and data types
- Conditional statements

- Loops and iterations
- Functions and modules
- Exception handling

Module 3:

Python for Data Analysis (Pandas, NumPy)

- Introduction to data analysis
- Reading and writing data files
- Data manipulation techniques
- Data cleaning and preprocessing

- Handling missing and duplicate values
- Data aggregation and transformation

Module 4:

Data Visualization & Reporting (Matplotlib and Seaborn)

- Data visualization principles
- Exploratory Data Analysis (EDA)
- Univariate and bivariate analysis
- Charts, graphs, and plots

- Trend and correlation analysis
- Business reporting
- Data storytelling

Module 5:

Power BI

- Building Dashboards and Reports Generation
- Data Analysis Expressions (DAX) and Power Queries
- Data manipulations

- Power Queries
- Creation of Measures and Calculated Columns

Module 6:

SQL & Database Management (MySQL)

- Database fundamentals
- Relational database concepts
- SQL queries and commands
- Filtering and sorting data

- Joins and subqueries
- Clauses and Data aggregation using SQL
- Basics of Triggers, store procedures

Module 7:

Exploratory Data Analysis (EDA)

- Understanding data structure
- Identifying patterns and trends
- Outlier detection
- Feature understanding
- Data preparation for machine learning

- Feature Encoding
- Feature Scaling
- Principal Component Analysis (PCA).

Module 8:

Machine Learning Fundamentals

- Introduction to machine learning
- Supervised learning concepts
- Regression models
- Classification models

- Unsupervised learning concepts
- Model evaluation techniques

Module 9:

Advanced Machine Learning

- Feature engineering
- Model tuning techniques
- Hyperparameter optimization

- Ensemble learning methods
- Handling imbalanced datasets

Module 10:

Artificial Intelligence

- Perceptron Models
- Activation function and Backpropagation
- Artificial Neural Networks (ANN)
- Convolution Neural Networks (CNN)

- Recurrent Neural Networks (RNN)
- Time series Analysis (AR, MA, ARIMA, SARIMA, SARIMAX)

Module 11:

Natural Language processing

- Text Mining and Text preprocessing
- Knowledge on NLP Terms [Corpus, corpora, bag of words, word2vec, embeddings etc.]

- Parts of speech tagging
- Sentimental analysis
- TF-IDF Vectorization.

Module 12:

Introduction to Generative AI

- Overview of Generative AI
- Large Language Models fundamentals
- How Gen AI models work

- Prompt engineering basics
- Responsible AI usage

Module 13:

Generative AI for Data Analytics

- AI-assisted data exploration
- Natural language to SQL
- Automated insights generation

- AI-powered reporting
- Productivity enhancement using Gen AI

Module 14:

Capstone Project & Industry Use Cases

- Real-world problem statement
- End-to-end data pipeline
- Analytics, ML, DL and Gen AI integration

- Business insights and presentation
- Deployment overview